nature electronics

ARTICLES

https://doi.org/10.1038/s41928-021-00689-4

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High-mobility p-channel wide-bandgap transistors based on hydrogen-terminated diamond/hexagonal boron nitride heterostructures

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Field-effect transistors made of wide-bandgap semiconductors can operate at high voltages, temperatures and frequencies with low energy losses, and are important for power and high-frequency electronics. However, wide-bandgap p-channel transistors perform poorly compared with n-channel ones, making complimentary circuits difficult to achieve. Hydrogen-terminated diamond is a potential p-type material for such devices, but surface transfer doping—thought to be required to generate conductivity—limits performance because it requires ionized surface acceptors that can lead to hole scattering. Here we show that p-channel wide-bandgap heterojunction field-effect transistors can be created, without surface transfer doping, using a hydrogen-terminated diamond channel and hexagonal boron nitride gate insulator. Despite having a reduced density of surface acceptors, the transistors have a low sheet resistance (1.4 kΩ) and large ON current (1,600 μm mA mm^{-1}) compared with other p-channel wide-bandgap transistors, due to a high room-temperature Hall mobility (680 cm^2 V^{-1} s^{-1}). The transistors also exhibit normally OFF behaviour with an ON/OFF ratio of 10^8.

Wide-bandgap semiconductors, such as silicon carbide (SiC) and gallium nitride (GaN), have advantages over silicon for power and high-frequency applications. The high breakdown electric field of wide-bandgap semiconductors can reduce the length of the drift region and increase its doping concentration in field-effect transistors (FETs), leading to reduced ON resistance and conduction loss. The wide bandgap also allows high-temperature operation, thereby reducing the size of cooling systems. As a result, SiC-based FETs are used in compact and low-loss power conversion systems. Due to their high carrier density and high carrier mobility, GaN-based high-electron-mobility transistors (HEMTs) are also used for radio-frequency power applications.

However, p-channel wide-bandgap FETs have poor performance compared with their n-channel counterparts due to low hole mobility, which leads to a high ON resistance and high conduction loss. It also limits the current-handling capabilities and degrades the high-frequency operation of FETs. This has constrained the production of complementary circuits with integrated n-channel and p-channel transistors, which would offer advantages such as low power consumption and simple driving circuitry[1]. Although electron mobilities in GaN HEMTs are 1,500–2,000 cm^2 V^{-1} s^{-1}, hole mobility in GaN-based p-channel FETs are below 30 cm^2 V^{-1} s^{-1} (refs. [1–8]). For SiC-based FETs, electron mobilities are 30–200 cm^2 V^{-1} s^{-1} (refs. [9,10]), whereas hole mobilities are lower than 17 cm^2 V^{-1} s^{-1} (refs. [1,11,12]). With gallium oxide (Ga_2O_3), producing any amount of p-type conductivity is a challenge[13].

Diamond is a wide-bandgap material that has a bulk phonon-limited intrinsic hole mobility higher than 2,000 cm^2 V^{-1} s^{-1} (refs. [14–17]), which makes it promising for use in high-mobility p-channel transistors. The high intrinsic mobility of diamond relates to its hardness: due to the high optical phonon frequency and high acoustic phonon velocity of diamond, the number of phonons that participate in carrier scattering is small[15,16]. Its other characteristics, such as a wide bandgap (5.47 eV), high breakdown field (>10 MV cm^{-1}) and high thermal conductivity (>2,200 W m^{-1} K^{-1}), also make diamond attractive for power applications[18,19], and these properties are superior to those of SiC, GaN and Ga_2O_3.

Diamond FETs typically use hydrogen-terminated diamond surfaces as a channel. Electron-acceptor materials, such as atmospheric adsorbates, acidic gases or high-work-function oxides, covering a hydrogen-terminated diamond surface capture electrons from the diamond valence bands and create holes at the surface. This surface transfer doping[20,21] induces p-type conductivity without relying on bulk dopants, and coating the diamond surface with an acceptor material is a common strategy for making hydrogen-terminated diamond FETs[18,22–28] (Supplementary Section A). However, this is accompanied by negatively charged acceptors near the diamond surface, which act as sources of carrier scattering, and reduce the carrier mobility[29,30]. The negative charges also cause a positive shift in the threshold voltage, leading to normally ON operation[22,24,25,27,28]: that is, a finite channel conductance at zero gate bias. This is undesirable, particularly in power electronics, for fail-safe operation and driving-circuit simplicity.

In this Article, we show that high-mobility p-channel wide-bandgap transistors can be made by reducing the density of surface acceptors on hydrogen-terminated diamond and without the need for surface transfer doping. Hexagonal boron nitride (hBN) and graphite are used as the gate insulator and electrode, respectively, and the transfer is made without exposing the diamond surface to air. The transistors exhibit a room-temperature mobility of 680 cm^2 V^{-1} s^{-1}, a sheet resistance of 1.4 kΩ and an ON current of 1,600 μm mA mm^{-1}, as well as normally OFF behaviour with an ON/OFF ratio exceeding 10^8. We also show that the output of our diamond FETs can be

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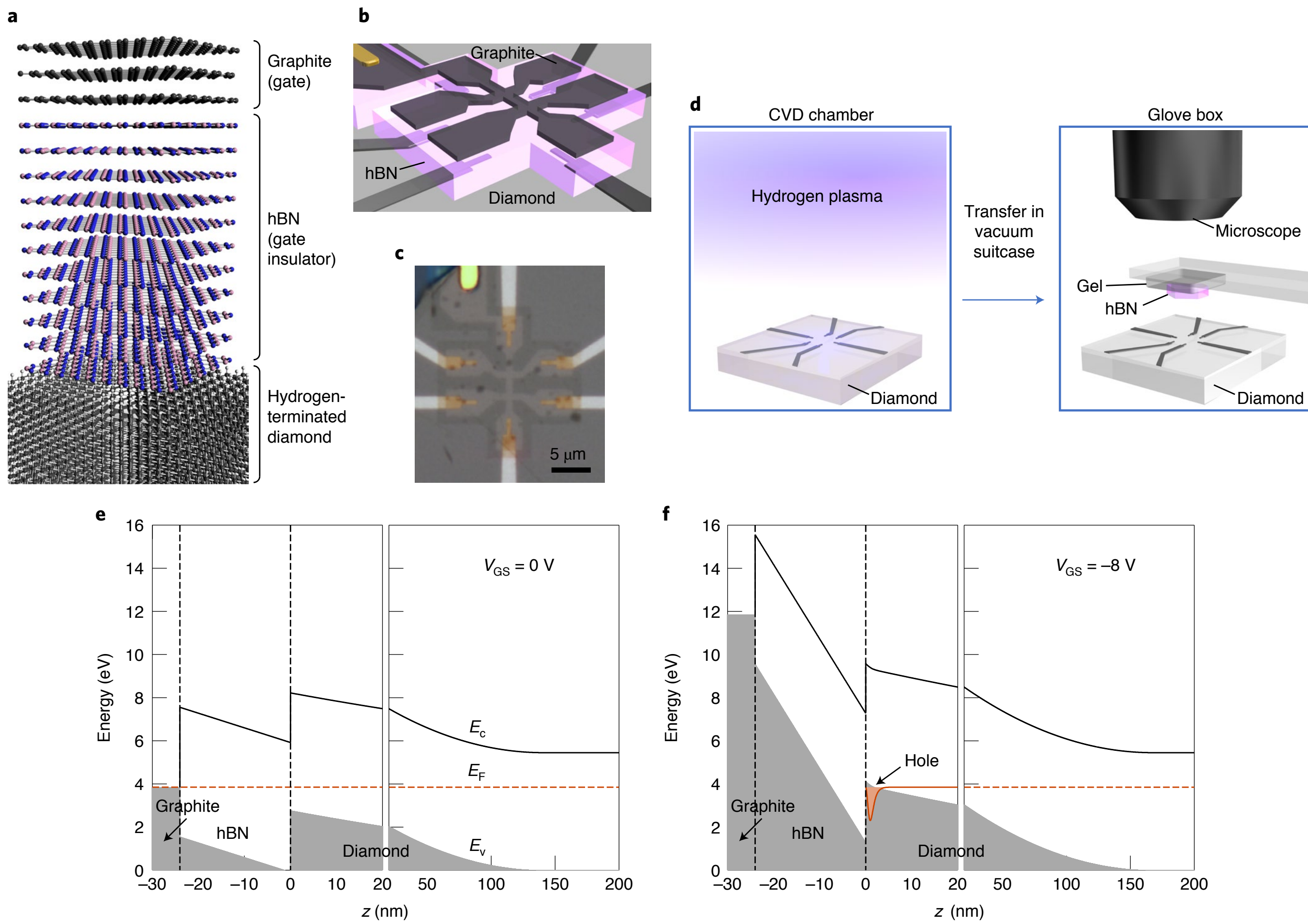

Fig. 1 | Diamond FET with hBN gate insulator and graphite gate. a, Schematic of the heterostructure consisting of graphite/hBN/hydrogen-terminated diamond. Here hBN is thicker in actual devices (24 nm, corresponding to 80 layers). **b**, Schematic of the diamond FET with hBN gate insulator and graphite gate. The diamond surface in the region covered with hBN is hydrogen terminated. The diamond surface in the other region is oxygen terminated, which makes the surface highly insulating and provides electrical isolation. **c**, Optical microscopy image of the fabricated device. **d**, Schematic of the transfer of diamond from the CVD chamber (where the diamond surface is hydrogenated by hydrogen plasma) to a glove box (where the diamond is laminated with an hBN thin crystal). The transfer is made in a vacuum suitcase without any exposure of the diamond surface to air. **e**,**f**, Energy-band diagrams of the graphite (gate; $z < -24$ nm)/hBN (-24 nm $< z < 0$)/diamond ($z > 0$) heterostructure for applied gate voltages of 0 V (**e**) and -8 V (**f**). The energy is relative to the valence band maximum deep inside the diamond. An inversion layer of holes is formed at the diamond surface for $V_{GS} = -8$ V. E_c, conduction band minimum; E_v, valence band maximum; and E_F, Fermi level.

explained on the basis of standard transport and transistor models. The presence of surface-acceptor states has been thought to be necessary for the surface conductivity of hydrogen-terminated diamond, but our results show that this is not the case and device performance can even be improved by their removal.

FETs based on hBN/diamond heterostructures. Schematic of the diamond FETs fabricated in this study is shown in Fig. 1a,b. A gated Hall-bar structure was used to evaluate the carrier density and mobility through Hall measurements (Fig. 1b,c). The gate insulator was produced by laminating a hydrogen-terminated diamond surface with a cleaved hBN single crystal, as in previous studies[31,32]. This method is useful for avoiding the formation of defects in the gate insulator, which occurs in conventional deposition techniques and can cause acceptor states. As a further step in this study, hBN lamination was conducted without exposing the hydrogen-terminated diamond surface to air (Fig. 1d), which greatly reduced the density of atmospheric acceptors. In addition, a thin graphite crystal was used as a gate electrode to reduce disorder at the interface between the gate and gate insulator[33]. Energy-band diagrams (Fig. 1e,f; Supplementary Section D provides details) obtained from a self-consistent solution of the Schrödinger and Poisson equations illustrate that an inversion layer of holes forms at the diamond surface under an applied gate voltage, without any surface transfer doping. Details of the device fabrication are given in Methods and Supplementary Fig. 3. In brief, after the deposition of Ti/Pt and annealing for forming ohmic contacts, the diamond surface was hydrogenated in a chemical vapour deposition (CVD) chamber. The diamond was directly transferred in a vacuum suitcase from the CVD chamber to an Ar-filled glove box, where the diamond was laminated with hBN and graphite by using the the Scotch-tape exfoliation and dry transfer techniques[34]. The subsequent processes were carried out outside the glove box, including the etching of graphite and hBN into a Hall-bar shape and the deposition of Ti/Au for leads and bonding pads.

Reduction in the density of atmospheric acceptors by this fabrication process was confirmed by measuring the surface conductivity of hydrogen-terminated diamond substrates (not laminated with hBN) by using an electrical prober system placed in the glove box. The surface conductivity was kept low when the diamond was

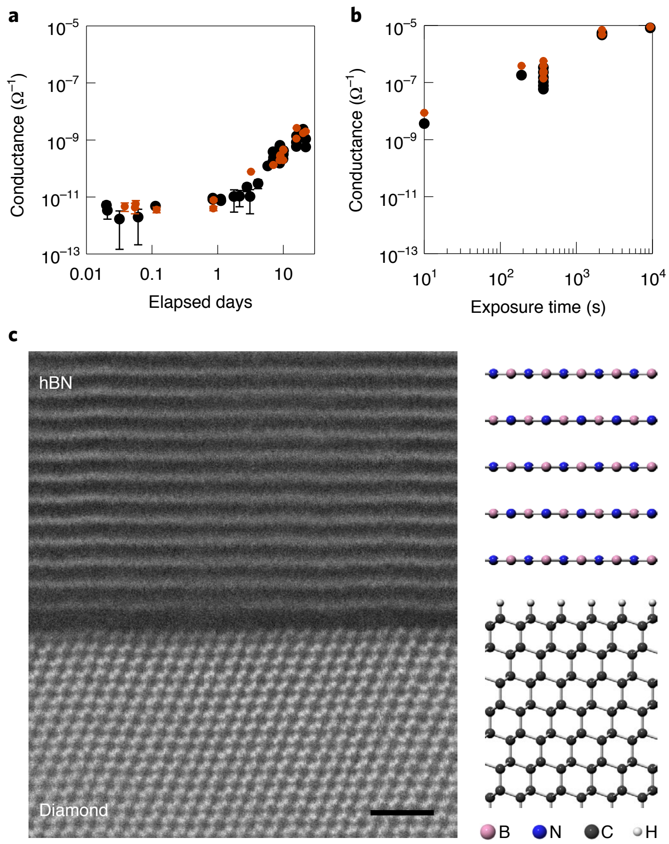

Fig. 2 | Reduction in acceptor density on hydrogen-terminated diamond with no exposure to air. **a**, Surface conductivity of hydrogen-terminated diamond kept in the glove box after vacuum transfer from the CVD chamber. The black and brown dots represent two different diagonal positions on the diamond substrate at which the prober needles are contacted. The error bars represent the standard error from linear fits of the current–voltage characteristics. **b**, Surface conductivity versus total air-exposure time. After measurements in which the diamond was kept in the glove box for about 20 days, as shown in **a**, a test was conducted to see the effect of exposure to air. The diamond was exposed to air for a certain time and brought back to the glove box, where the surface conductivity was measured. This procedure was repeated as the exposure time was increased. The surface conductivity is plotted against cumulative total time. **c**, STEM image of an hBN/hydrogen-terminated (111) diamond heterostructure prepared with the air-free process. The image was taken from the $[\bar{1}10]$ crystallographic direction of diamond. The hydrogen atoms are invisible in the STEM image. Scale bar, 1 nm.

transferred from the CVD chamber and stored in the glove box. The resistance was kept higher than ~10^{11} Ω for at least three days (Fig. 2a). Note that the hBN lamination for fabricating FETs was completed within 3 h of the diamond transfer from the CVD chamber to the glove box. The gradual increase in conductivity may be due to the adsorption of remaining atmospheric gases in the glove box. When the diamond surface was exposed to air, it turned into a conductive state with a resistance of ~100 kΩ, which is expected because the hydrated ions in atmospheric adsorbates act as surface acceptors[20,21] (Fig. 2b). The increased conductivity rules out the possibility that the low conductivity (Fig. 2a) was caused by the failure of hydrogenation of the diamond surface. These observations indicate that the atmospheric acceptors that cause the surface conductivity can be substantially reduced by using the above fabrication process. This was further evidenced by the normally OFF behaviour of the FETs described below. A scanning transmission electron microscopy (STEM) image of an hBN/diamond heterostructure prepared with the air-free process (Fig. 2c) indicates that the distance between the first hBN layer and hydrogen-terminated diamond surface is nearly the same as the interlayer spacing (0.333 nm) of hBN (hydrogen atoms are invisible in the STEM image and the C–H bond length is estimated to be 0.112 nm for the hydrogen-terminated (111) diamond surfaces[35]). The STEM image of a larger-area hBN/diamond heterostructure (Supplementary Fig. 5a) shows that the spacing between the first hBN layer and diamond is uniform over a large area. These images also suggest a low density of contaminants at the interface.

Room-temperature characteristics of hBN/diamond FETs. The electrical properties of our FETs were evaluated by making conductivity and Hall-effect measurements. Figure 3a shows the gate-voltage dependence of Hall mobility at room temperature. The sign of the Hall voltage is positive, indicating that the charge carriers are holes. The Hall mobility reaches 680 $cm^2 V^{-1} s^{-1}$ at a gate voltage $V_{GS}=-10$ V. This is higher than those of FETs fabricated with air-exposed hydrogen-terminated diamond and hBN (ref. [31]), indicating that the reduction in surface-acceptor density is effective in improving mobility. Moreover, this room-temperature mobility is one of the highest ever reported for p-channel FETs based on a diamond surface channel or any other wide-bandgap semiconductor. The effective mobility $\mu_{\rm eff}=\frac{t_{\rm hBN}}{\epsilon_{\rm hBN}}\frac{\sigma}{|V_{\rm GS}-V_{\rm th}|}$ and field-effect mobility $\mu_{\rm FE}=\frac{t_{\rm hBN}}{\epsilon_{\rm hBN}}\left|\frac{\partial\sigma}{\partial V_{\rm GS}}\right|$ ($t_{\rm hBN}$ and $\epsilon_{\rm hBN}$ are the thickness and dielectric constant of hBN, respectively; σ, sheet conductance in the linear regime; V_{th}, threshold voltage) were also evaluated from the transfer curve (Fig. 3b). If we use $\epsilon_{\rm hBN}/\epsilon_0=3.2$ (ϵ_0 is the vacuum permittivity)[36,37], the gate-voltage dependence of the effective mobility is in good agreement with that of Hall mobility (Fig. 3a). The field-effect mobility is larger than the effective mobility, especially at small gate voltages. This is because $\mu_{\rm FE}=\mu_{\rm eff}+(V_{\rm GS}-V_{\rm th})\frac{\partial\mu_{\rm eff}}{\partial V_{\rm GS}}$ and the sign of $(V_{\rm GS}-V_{\rm th})\frac{\partial\mu_{\rm eff}}{\partial V_{\rm GS}}$ is positive (Supplementary Section F).

Normally OFF operation of our FET was clearly indicated by the transfer curve and V_{GS} dependence of the Hall carrier density ($p_{\rm Hall}$) at room temperature (Fig. 3c). A linear fit to the $p_{\rm Hall}-V_{\rm GS}$ curve gives a threshold voltage V_{th} of −0.99 V. This threshold voltage is close to the value predicted by the standard formula using device parameters such as the work-function difference between hydrogen-terminated diamond and graphite, as described below. The carrier density reaches 6.6×10^{12} cm^{-2} at $V_{GS}=-10$ V. At this gate voltage, the sheet conductance is 7.2×10^{-4} Ω^{-1}, which is equivalent to a sheet resistance of 1.4 kΩ. This ON sheet resistance is one of the lowest ever reported for diamond FETs. Since the ON resistance is usually sacrificed for normally OFF operation[38–41], the low ON resistance compatible with normally OFF operation in our FET is remarkable. This resistance is even comparable to the lowest sheet resistances reported for hydrogen-terminated diamond exposed to NO_2 or coated with WO_3 or V_2O_5 with no FET structure[25,42,43]. The low sheet resistance of NO_2-exposed or WO_3- or V_2O_5-coated hydrogen-terminated diamond is due to a high carrier density of the order of 10^{14} cm^{-2}, whereas the low sheet resistance of our FET is mainly due to high carrier mobility.

The output characteristics (Fig. 3d) also indicate the excellent performance of our FET. The output curves display the near-ideal saturation in drain current. The output currents (Fig. 3e) can be quantitatively explained on the basis of a long-channel transistor model (Supplementary Section E provides the detailed modelling). The maximum current normalized by the channel width ($I_D^{\rm max}/W_G$) is 200 mA mm^{-1}. This value is again exceptionally high among those of normally OFF diamond FETs[39,44–46], despite the long gate length L_G^* of 8.09 μm (Supplementary Table 1 lists the device dimensions). Note that the maximum current is generally inversely proportional

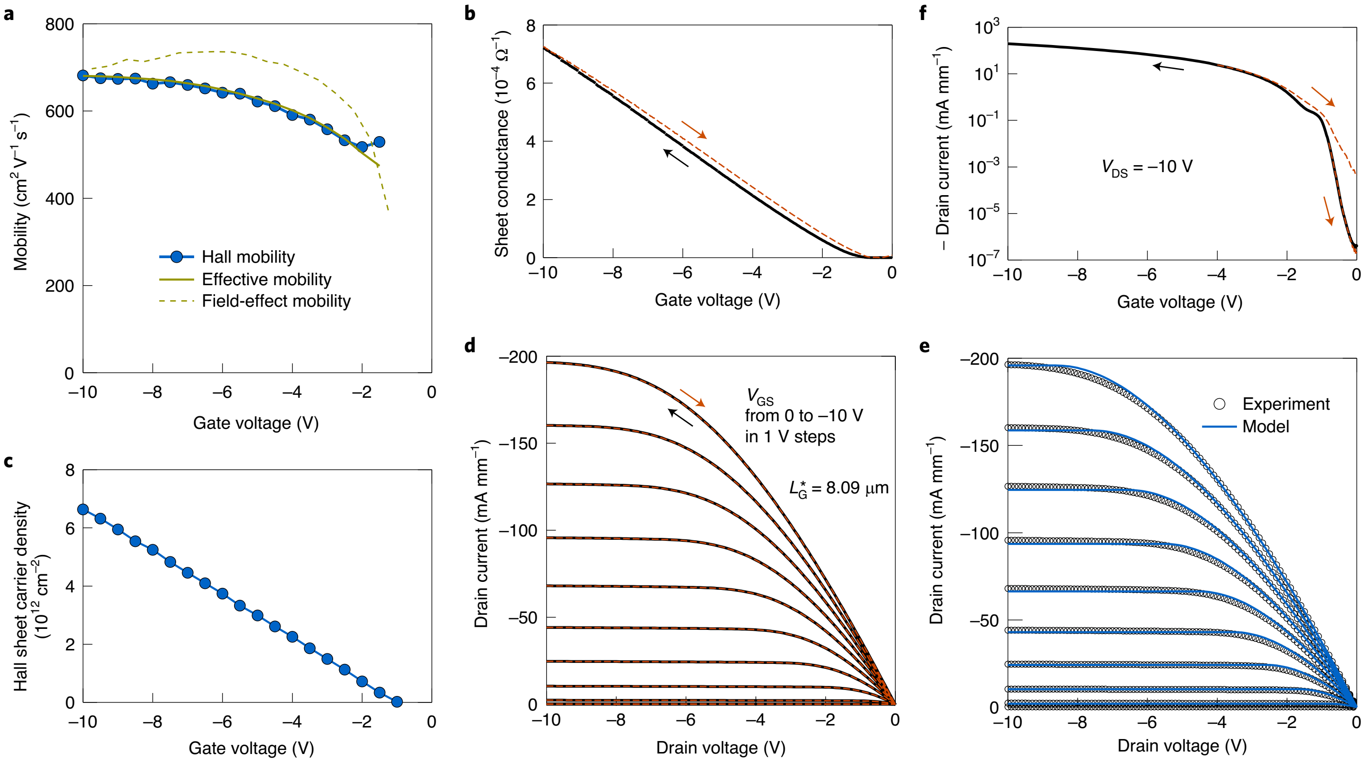

Fig. 3 | Electrical characteristics of diamond FET at room temperature. **a**, Hall, effective and field-effect mobilities as a function of gate voltage. A relative dielectric constant value of 3.2 (refs. [36,37]) for hBN was used for calculating the effective and field-effect mobilities. **b**, Transfer characteristics in the linear region. The channel sheet conductance was measured with the four-probe configuration. The solid and dashed lines were obtained as the gate voltage was swept from 0 to -10 V and from -10 to 0 V, respectively. The hysteresis, in which the magnitude increased with the maximum applied gate voltage, may be due to the relocation of residual surface acceptors under electric fields. **c**, Hall carrier density as a function of gate voltage. **d**, Output characteristics. The drain current measured with the two-probe configuration is plotted as a function of drain voltage for different gate voltages. **e**, Measured output characteristics (circles; same as the black lines in **d**) and calculated ones (lines) using a charge sheet model (Supplementary Section E provides the detailed modelling). **f**, Drain current as a function of gate voltage at a drain voltage of -10 V. The solid line corresponds to a gate sweep from 0 to -10 V. The two dashed lines correspond to gate sweeps from -0.8 to 0 V and -4 to 0 V.

to the gate length. The gate-length-normalized maximum current ($L_G^* I_D^{max}/W_G$) is 1,600 μm mA mm^{-1}. The transconductance is also high, reaching 37 mS mm^{-1} (Supplementary Fig. 6a). These high ON-state characteristics are the result of the high carrier mobility achieved by the reduced surface-acceptor density.

Despite the high ON-state characteristics, the FETs can be switched off with a low source–drain leakage current. The gate-voltage dependence of the drain current in the saturation region (drain voltage, $V_{DS} = -10$ V) gives an ON/OFF ratio larger than 10^8 (Fig. 3f). The leakage current in the OFF state ($V_{GS} = 0$ V) is almost at the noise level of our measurement setup. The drain current shows hysteresis and its magnitude increases with the maximum applied gate voltage. The hysteresis may be caused by the relocation of the residual surface acceptors under electric fields. The minimum subthreshold swing SS is 130 mV dec^{-1} at $V_{GS} = -0.55$ V. The interface trap density D_{it} can be evaluated using the equation[47]

$$SS = \ln 10 \frac{dV_{GS}}{d(\ln I_D)} = \ln 10 \frac{k_B T}{e} \frac{C_{hBN} + C_{depl} + e^2 D_{it}}{C_{hBN}}, \quad (1)$$

where k_B is the Boltzmann constant; T, the absolute temperature; e, the elementary charge; C_{hBN} (ϵ_{hBN}/t_{hBN}), the capacitance of hBN gate insulator; and C_{depl}, the capacitance of the diamond depletion layer. If we use $t_{hBN} = 24$ nm and $\epsilon_{hBN}/\epsilon_0 = 3.2$, then $C_{hBN} = 0.12\,\mu F\,cm^{-2}$. Here C_{depl} is estimated to be $3.0\times10^{-2}\,\mu F\,cm^{-2}$ from the calculated depletion-layer thickness for $N_D = 500$ ppb and $N_A = 7$ ppb, where N_D and N_A are donor (nitrogen) and acceptor (boron) concentrations obtained from a secondary ion mass spectrometry measurement. Using these values, D_{it} is evaluated to be $6.8\times10^{11}\,cm^{-2}\,eV^{-1}$. This interface state density is lower than most of the values reported for diamond/insulator interfaces in recent studies[48–50]. This is presumably due to the dangling-bond-free surface of hBN and reduced adsorbate density. The gate leakage current is almost at the noise level (5×10^{-13} A) for gate voltages between 0 and −10 V. This current corresponds to $3\times10^{-7}\,A\,cm^{-2}$ if it is normalized by the total area of the graphite gate electrode. If it is normalized by the channel width, it corresponds to $5\times10^{-7}\,mA\,mm^{-1}$ (Supplementary Fig. 6b). The gate leakage current is sufficiently small in both representations. The gate–source breakdown field is at least higher than 4.2 MV cm^{-1} (10 V/24 nm).

Temperature-dependent properties of hBN/diamond FETs. Temperature dependence of the electrical properties gives another insight into the quality of our FET channel (Fig. 4). Generally, the mobility of carriers induced by surface acceptors on a hydrogen-terminated surface decreases as the temperature falls below room temperature[25,51]. This is because negative charges near the surface are the dominant source of carrier scattering that determines mobility. Lowering the temperature increases the population of lower-energy holes, which are influenced more by potential fluctuations. The mobility of our FETs clearly increases with decreasing temperature from 300 to 150 K. The mobility exceeds 1,000 cm^2 V^{-1} s^{-1} at 150 K for $V_{GS} = -8$ V (Fig. 4a). The negative sign of the derivative of mobility with respect to

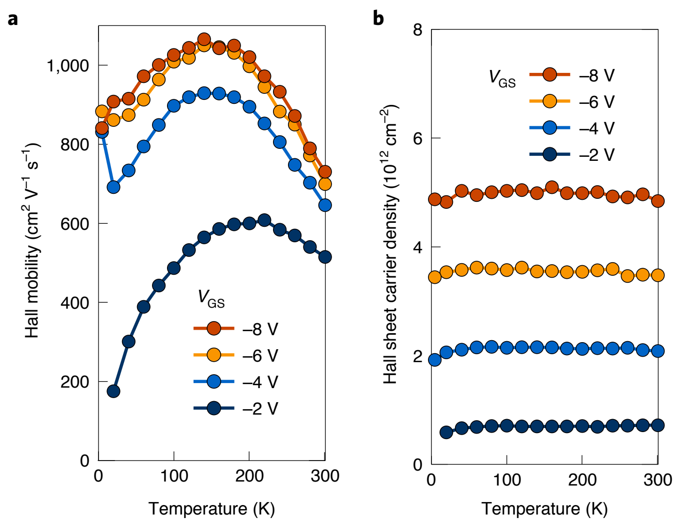

Fig. 4 | Temperature-dependent electrical characteristics of diamond FET. **a**,**b**, Temperature dependence of Hall mobility (**a**) and Hall carrier density (**b**) at different gate voltages.

temperature indicates the contribution of phonon scattering, which is now clearly observable in the temperature range of 150–300 K due to the suppression of ionized impurity scattering (Supplementary Section C and Supplementary Fig. 12a provide the detailed modelling of mobility). In addition, high mobility persists at a low temperature, even for a low carrier density of 2×10^{12} cm^{-2} (Fig. 4a,b). This carrier density is lower than those required to maintain high mobility in previous studies[31,32,52]. Furthermore, the carrier density (Fig. 4b) is nearly independent of temperature down to the lowest temperature (4.5 K), indicating that freezing out of the carriers does not occur even at such a low temperature. All these observations indicate an improvement in channel quality and demonstrate that surface acceptors are unnecessary for obtaining highly conductive states in hydrogen-terminated diamond. Gate controllability is retained even at cryogenic temperatures (Supplementary Fig. 7), which may be used to fabricate a device that works in an environment with large temperature variations, for example, in space.

Mobility and normally OFF property of hBN/diamond FETs. Figure 5 benchmarks the performance of our FETs at room temperature against those in the literature. The mobility is one of the highest among reports of FETs based on a diamond surface channel (Fig. 5a). It is also higher than p-channel GaN-based heterostructures[2–8] ($\leq$30 cm^2 V^{-1} s^{-1}) and p-channel SiC-based FETs[11,12] (7–17 cm^2 V^{-1} s^{-1}) (the mobilities for SiC are not shown in Fig. 5a because the carrier densities are not reported in refs. [11,12]). High mobility lowers the channel sheet resistance (approaching 1 kΩ) and increases the ON current I_{ON} (I_D^{max}/W_G). The gate-length-normalized ON current $I_{ON}L_G^*$ is one of the highest among the reports of p-channel FETs made of diamond and other wide-bandgap semiconductors (Fig. 5b). Importantly, normally OFF behaviour with a large ON/OFF ratio is simultaneously obtained.

The normally OFF behaviour of our FETs can be understood by considering that the densities of atmospheric adsorbates and the resulting negative charges are reduced in the FETs. The threshold voltage is given by[47]

$$V_{th} = -\frac{e(n_{depl} - n_{ic})t_{hBN}}{\epsilon_{hBN}} + \psi_s(p_{2D} \to 0) + \phi_{ms}. \qquad (2)$$

Here n_{depl} is the sheet density of the fixed charge in the depletion layer and n_{ic} is the net sheet density of the interface negative charge. Also, $\psi_s(p_{2D}\to 0)$ (<0) is the surface potential in the limit of low hole sheet density p_{2D} and $e\phi_{ms} = e\phi_m - e\phi_s$ is the difference between the work function ($e\phi_m$) of the graphite gate and that ($e\phi_s$) of hydrogen-terminated diamond. This equation indicates that a decrease in the interface negative charge causes a negative shift in threshold voltage V_{th}. The threshold voltage calculated using the above equation is in good agreement with the measured threshold voltage (Supplementary Section D and Supplementary Fig. 12c). The increase in donor (for example, nitrogen) density in diamond increases n_{depl} and causes a negative shift in V_{th}, but this enhances undesirable ionized impurity scattering. A partial ultraviolet ozone treatment[38,39] achieves normally OFF operation, presumably by increasing the spatially averaged value of $e\phi_s$, but non-uniform hydrogen termination should cause local potential fluctuations that result in carrier scattering. Introducing positive charges into the gate insulator to compensate the interface negative charge[40] should also cause local potential fluctuations. By contrast, the normally OFF operation in our FETs is based on a reduction in interface negative charge, which decreases carrier scattering and thus enhances mobility. Note that the choice of a gate material with an appropriate work function $e\phi_m$ would enable the threshold voltage to be tuned without lowering the mobility.

The holes in our FETs are generated in the absence of acceptors by an upward band bending due to gate bias (Fig. 1f), a mechanism equivalent to the formation of an inversion channel in Si metal–oxide–semiconductor FETs, where electrons are generated in the absence of donors. In fact, the output characteristics are in excellent agreement with those predicted by the standard model for an inversion channel (Fig. 3e). The positive charge of holes is mostly balanced by the negative charge of electrons in the gate electrode. This remote, uniform negative charge does not scatter holes and leads to an increase in mobility. This avenue for increasing the mobility is analogous to that available in GaAs HEMTs in which a doped layer is spatially separated from the channel of the transistors. Although the doped layer is required to compensate for the high density of surface states in GaAs HEMTs[53], remote dopants are unnecessary in our FETs due to the low density of surface states in hBN. The reason for using the hydrogen-terminated surface is not the surface transfer doping phenomenon but its shallow valence bands, which are rooted in the presence of $C^{\delta-}$–$H^{\delta+}$ dipoles[35], low density of surface states and availability of good ohmic contacts (the last point relates to the first two).

Despite the improved transport characteristics, the decrease in mobility at a low temperature (Fig. 4a) and increase in mobility with carrier density (Fig. 5a) suggest that there is still some contribution from charged impurity scattering (Supplementary Section C and Supplementary Fig. 12a,b provide the modelling of mobility). The charge that causes carrier scattering may arise from atmospheric gas that was not sufficiently removed by the purifier of the glove box and gets adsorbed on the diamond surface. Alternatively, charge may have become trapped in the interface states due to defects on the surface of diamond or hBN. The interface state density estimated from the subthreshold swing is 6.8×10^{11} cm^{-2} eV^{-1}, as described earlier. A plausible origin of the interface states is defects in diamond or hBN near the interface, such as broken bonds or impurities. The defects may be imperfections of hydrogen termination (for example, dangling bonds or a different kind of termination such as oxygen), lattice distortions or carbon vacancies in diamond. Boron or nitrogen vacancies and carbon impurities may be present in hBN (ref. [54]). These defects possibly act as interface states of the donor, acceptor or amphoteric type, depending on their kind. For instance, a C=C bond formed near the surface of diamond has been suggested to act as an acceptor state[55]. A carbon dangling bond (missing H termination) at the diamond surface may act as

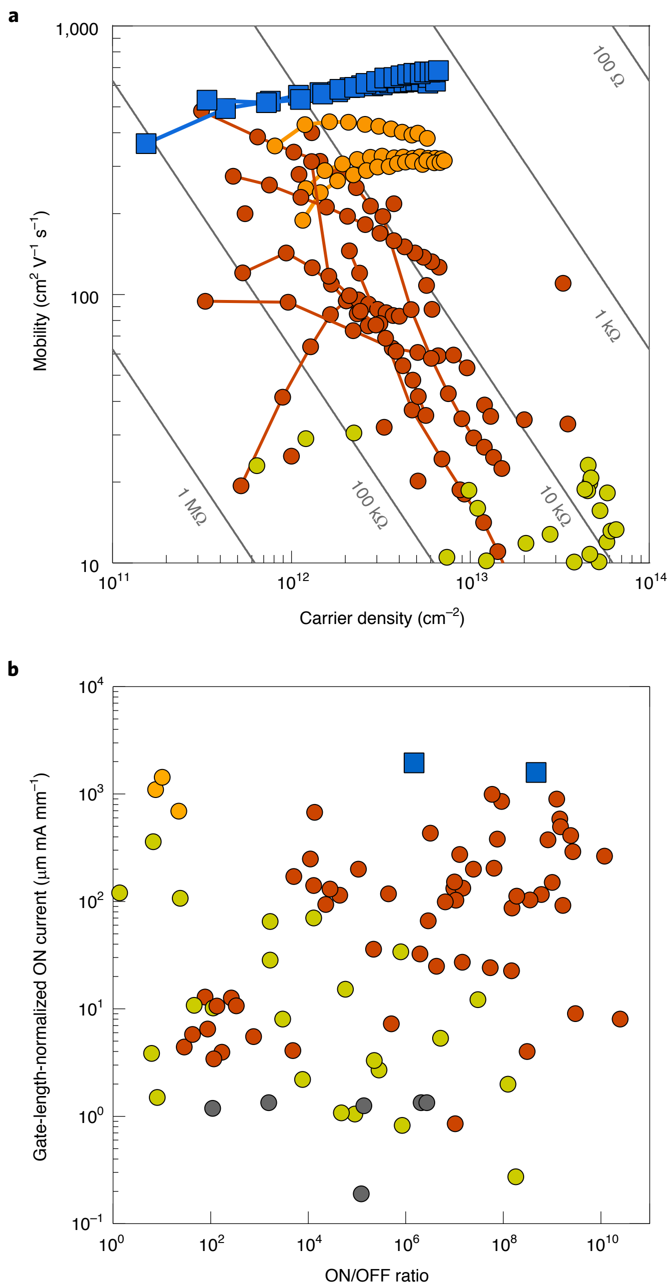

Fig. 5 | Comparison of room-temperature performances of diamond FET and earlier works. a, Mobility and carrier density of our present FETs (blue squares; two devices), other diamond FETs (brown and orange dots; orange dots, previous FETs fabricated with air-exposed hydrogen-terminated diamond and hBN) and p-channel GaN-based heterostructures (green dots). References for the data are given in Supplementary Section B. The mobility-extraction methods for the diamond FETs are tabulated in Supplementary Table 2. The lines indicate the sheet resistances of 100 Ω to 1 MΩ. **b**, Normalized ON current and ON/OFF ratio of our present FETs (blue squares; two devices), other diamond FETs (brown and orange dots; orange dots, our previous FETs fabricated with air-exposed hydrogen-terminated diamond and hBN), p-channel GaN FETs (green dots) and p-channel SiC FETs (grey dots). References for the data are given in Supplementary Section B.

a donor state near the valence band, like a silicon dangling bond (known as the P_b centre) at the Si/SiO_2 interface[50]. Such donor-type interface states should be positively charged and reduce mobility. Acceptor-type interface states should have less of an effect on mobility than donor-type interface states, because they are neutral when the Fermi level is lower than their energy levels, which should be the case particularly when the carrier density is high. Differences in mobility and threshold voltage between the two devices fabricated in this study, which seem to relate to the different environmental conditions for the two devices before the hBN lamination (Methods), suggest that some adsorbates still exist and are causes of the reductions in both mobility and threshold voltage. Indeed, a detailed analysis indicates that interface negative charges due to the remaining adsorbates mainly caused the reduction in mobility, whereas positive charges of donor-type interface states additionally contributed to it (Supplementary Section D). A further reduction in atmospheric adsorbates and preparation of an interface with a lower density of structural defects, for example, using an atomically flat diamond[52,56], will increase mobility.

Conclusions

We have shown that hydrogen-terminated diamond/hBN heterostructures can be used to fabricate high-mobility p-channel wide-bandgap FETs. Although previous studies on hydrogen-terminated diamond FETs introduced surface acceptors, it was found that these were unnecessary to exhibit hole transport; in fact, higher hole mobility, improved ON-state characteristics and normally OFF operation can be achieved without them. Reducing defects would also increase device stability and reliability, which are important requirements for practical applications. Our p-channel wide-bandgap FETs exhibited a room-temperature mobility of $680\,cm^2\,V^{-1}\,s^{-1}$, sheet resistance of 1.4 kΩ and ON current of 1,600 μm mA mm^{-1}. These values illustrate the potential of diamond FETs for power and high-frequency electronics applications. The hole mobility in our FET increases with decreasing temperature from 300 to 150 K and exceeds $1,000\,cm^2\,V^{-1}\,s^{-1}$ at 150 K. The contribution of phonon scattering to mobility in this temperature range, which had been masked by ionized impurity scattering in previous studies, is now clearer. Since the phonon-limited hole mobility is higher than $2,000\,cm^2\,V^{-1}\,s^{-1}$ at room temperature in bulk diamond[14–17], further improvements in the mobility of diamond FETs should be possible[30,57]. This could be achieved by reducing extrinsic and intrinsic defects near the diamond/gate insulator interface.

The FET characteristics were quantitatively explained on the basis of standard FET models, which provides a solid basis for the further development of diamond FETs. A promising direction for this development would be a complementary circuit in which a p-channel diamond FET and n-channel GaN-based FET are integrated[1,23]. High-frequency, high-output diamond FETs with an intrinsic diamond heat spreader could also be developed, which use, for example, the non-surface-transfer-doped diamond/hBN heterostructure in the channel region and use surface transfer doping in the access region. Our approach for fabricating diamond transistors could thus lead to the development of high-performance wide-bandgap p-channel transistors and complementary circuits.

Methods

Device fabrication. The FETs were fabricated on IIa-type (111) single-crystalline diamonds synthesized in a high-temperature, high-pressure process (purchased from Technological Institute for Superhard and Novel Carbon Materials). The (111) orientation was chosen because of its tendency to have higher surface conductivity than the (001) orientation[58,59] and the small lattice mismatch between hBN and hydrogen-terminated (111) diamond surface[31]. Experimental results for two devices (C1 and C2) are described in this paper. The dimensions of the diamonds were 2.0 mm × 2.0 mm × 0.4 mm for device C1 and 2.5 mm × 2.5 mm × 0.3 mm for device C2. A secondary ion mass spectrometry measurement on a purchased diamond substrate with the same quality specifications indicated that the concentration of nitrogen was 0.5 ppm ($8\times10^{16}\,cm^{-3}$) and that of boron was 7 ppb ($1\times10^{15}\,cm^{-3}$). The surfaces of the diamonds were polished and cleaned in hydrofluoric acid at room temperature and in a mixture of sulfuric and nitric acid at 200 °C before device fabrication.

Hall-bar electrodes were produced using electron-beam lithography and deposition of Ti/Pt (5 nm/5 nm) (Supplementary Fig. 3a). The diamond was annealed in H_2 gas at a temperature of 650 °C for 35 min and exposed to hydrogen plasma at 600 °C for 10−12 min in a microwave-plasma-assisted CVD chamber (Seki Technotron, AX5200-S). The diamond was then annealed in H_2 gas at 650−710 °C for 35 min and exposed to hydrogen plasma at 600−670 °C for 10 min in another CVD chamber (Seki Technotron, AX5000). The flow rate of H_2 gas was 500 s.c.c.m. These processes resulted in the formation of TiC ohmic contacts and hydrogenation of the diamond surface (Supplementary Fig. 3b). The diamond was transferred from the second CVD chamber to an Ar-filled glove box in a custom-made vacuum suitcase without exposing the diamond surface to air. Within 30 h before diamond transfer, single-crystalline hBN was cleaved in the glove box using Scotch tape and transferred onto a polydimethylsiloxane sheet (Gel-Pak, PF-60-X4) on an acrylic plate. The diamond surface was laminated with the cleaved hBN thin crystal under an optical microscope in the glove box by using a dry transfer technique[34] (Supplementary Fig. 3c). The lamination was completed within 3 h after the diamond transfer from the CVD chamber to the glove box. The sample was then annealed at 300 °C in Ar in the glove box for 3 h (Supplementary Fig. 3d). A graphite crystal (Kish graphite, CoorsTek) for the gate electrode was transferred onto hBN in a similar manner, followed by annealing at 300 °C for 1 h (Supplementary Fig. 3e,f). The sample was then taken out of the glove box. The thickness of the hBN and graphite was measured using atomic force microscopy (Hitachi High-Tech, AFM5100N, for device C1; L-trace II, for device C2) in the tapping mode by using a silicon tip. The thickness of the hBN was 24 nm for devices C1 and C2 (Supplementary Fig. 2). The thickness of graphite was 6 nm for device C1 and 20 nm for device C2. The graphite and hBN were etched into a Hall-bar shape using the plasma generated from a mixture of N_2, O_2 and CHF_3 gases with flow rates of 96, 2 and 2 s.c.c.m. at a total pressure of 10 Pa (Supplementary Fig. 3g,h). The etch rates of graphite and hBN were ≥7 and ≥16 nm min^{-1} for a radio-frequency power of 35 W. This process also converted the diamond surface, except the region under hBN, into an oxygen-terminated one, which was used for device isolation. Another flake of hBN was placed (Supplementary Fig. 3i) so that the gate lead deposited on it did not touch the hydrogen-terminated surface at the edge of the etched hBN, which could otherwise have resulted in gate leakage. The thickness of hBN was ~73 nm for device C1 and ~63 nm for device C2. Finally, Ti/Au (10 nm/100 nm) was deposited for leads and bonding pads (Supplementary Fig. 3j).

The vacuum suitcase used for the fabrication of device C1 was equipped with a non-evaporated getter pump and the pressure was lower than 10^{-7} torr during the transfer. The diamonds for device C2 and STEM specimen were transferred using the same suitcase, but in this case, it was just vacuum sealed with no pump used during the transfer. The pressure during the transfer was lower than 3×10^{-5} torr in this case. The Ar gas in the glove box was circulated via a purifier consisting of molecular sieves and heated copper to maintain the purity of Ar gas with the oxygen content below 0.5 ppm and water content below 2 ppm. The fabrication of device C1 used another external gas purifier (NuPure Optics, Model 4000 OA), which could remove acid gases and was inserted between the outlet of the purifier (described above) and the inlet of the glove box. The supplemental gas for controlling the inner pressure of the glove box was Ar with a purity of 99.9999% provided from a cylinder.

Measurement setup. The devices were set in a cryogen-free cryostat with a superconducting magnet. The transfer and Hall characteristics were measured with a gated Hall-bar configuration using a source measure unit (Keysight Technologies, B2901A) for biasing the gate, a function generator (Agilent Technologies, 33220A) for biasing the drain, two voltage amplifiers (DL Instruments, 1201) for measuring the longitudinal and Hall voltages and a current preamplifier (DL Instruments, 1211) for measuring the drain current. The measurements were performed by feeding a small positive and negative current, namely, I_{D+} and I_{D-}, respectively, in the linear regime (by applying a positive and negative voltage through a 10 MΩ series resistor). The absolute value of the applied voltage was smaller than 300 mV; therefore, the drain current was less than 30 nA, corresponding to ≤(3.4−3.5) × 10^{-2} mA mm^{-1}. Sheet resistance ρ was calculated as $\rho = (V_{p+} - V_{p-})/(I_{D+} - I_{D-})(W_G/L_p)$. Here V_{p+} and V_{p-} are the voltages between the longitudinal voltage probes of the Hall bar for the positive and negative currents, respectively; L_p is the distance between the voltage probes; and W_G is the gate width. Sheet conductance σ was obtained as $1/\rho$. A magnetic-field sweep between −1 and +1 T was used for the Hall measurements. Hall coefficient R_H was obtained from a linear fit of the magnetic-field dependence of Hall resistance $(V_{H+} - V_{H-})/(I_{D+} - I_{D-})$, where V_{H+} and V_{H-} are the voltages between the Hall probes of the Hall bar for positive and negative currents, respectively. The Hall carrier density and Hall mobility were obtained as $p_{Hall} = 1/(eR_H)$ and $\mu_{Hall} = R_H\sigma$. True carrier density p and drift mobility μ are given by $p = \gamma_H p_{Hall}$ and $\mu = \mu_{Hall}/\gamma_H$, where γ_H is the Hall scattering factor, which depends on the scattering mechanism and degree of warping of the valence bands. The Hall scattering factor has been measured and calculated to be around 0.8 for p-type bulk diamond at 300 K (ref. [60]). Since no estimates of γ_H have been made for the hole gas confined at the diamond surface, we assumed it to be unity. The output characteristics were measured with a two-point configuration using the source measure unit for biasing the gate, the function generator for biasing the drain and the current preamplifier for measuring the drain current. The gate leakage current was separately measured by applying a gate voltage and measuring the current through the drain and source by using the current preamplifier. For the electrical measurements of the hydrogen-terminated diamond surface in the glove box, prober needles made of Au-based alloy and with a tip diameter of 20 μm were directly contacted at diagonal positions of the surface of a hydrogen-terminated diamond substrate (2.6 × 2.6 × 0.3 mm IIa-type (111) single-crystalline diamond synthesized by CVD, purchased from Element Six; nominal nitrogen concentration below 1 ppm and boron concentration below 0.05 ppm).

Sample preparation for TEM. An atomically flat diamond surface was formed on a mesa structure on a Ib-type (111) single-crystalline diamond substrate (purchased from Sumitomo Electric Industries) using microwave-plasma-assisted CVD with a low CH_4/H_2 ratio (0.025%)[56]. The microwave power, pressure and stage temperature were 800 W, 80 torr and 810−820 °C, respectively. The flow rate of H_2 gas was 1,000 s.c.c.m. Atomic force microscopy measurements confirmed the terrace widths to be ~1 μm. The diamond surface was exposed to hydrogen plasma, transferred to the glove box using a vacuum suitcase and laminated with a flake of hBN in the glove box, in the same manner as the device fabrication. The hBN/diamond stack was covered with ~30-nm-thick carbon using a carbon evaporation coater (JEOL, JEC-560) for protection and preventing charge up. For protection against milling, ~1-μm-thick carbon was deposited on the hBN/diamond stack in the shape of a 12 × 2 μm^2 strip with the long side parallel to the $[\bar{1}\bar{1}2]$ direction of the diamond using a focused ion beam system (JEOL, JIB-4000). The deposition was performed with a Ga^+ ion beam (10 kV and 50 pA) under phenanthrene gas injection. Then, ~1-μm-thick carbon was further deposited in the shape of a 30 × 3 μm^2 strip covering the 12 × 2 μm^2 strip with Ga^+ (30 kV and 200 pA). In the focused ion beam system, trenches were etched around the larger strip with Ga^+ (30 kV and 30 nA to 700 pA) to cut out a cross-sectional slice with dimensions of ~25 μm × 2 μm × 5 μm. The slice was picked up and fixed on a copper grid with epoxy. The slice was further thinned with Ga^+ (30 kV and 700 to100 pA) to a thickness of ~80 nm. Finally, gentle milling with Ga^+ (5 kV and 30 pA) was performed to remove surface damage.

TEM imaging and analysis. TEM imaging was carried out using a Cs-corrected TEM (JEOL, JEM-ARM200F with a cold field-emission gun) operated at 200 kV. The $[\bar{1}10]$ crystallographic direction of diamond was adjusted to the electron-beam direction. A Fourier transform was performed using the DigitalMicrograph software (Gatan).

Data availability

Source data are provided with this paper. All other data that support the plots within this paper and other findings of this study are available from the corresponding author upon reasonable request.

Received: 8 April 2021; Accepted: 10 November 2021;
Published online: 23 December 2021

References

1. Bader, S. J. et al. Prospects for wide bandgap and ultrawide bandgap CMOS devices. *IEEE Trans. Electron Devices* **67**, 4010–4020 (2020).
2. Reuters, B. et al. Fabrication of p-channel heterostructure field effect transistors with polarization-induced two-dimensional hole gases at metal–polar GaN/AlInGaN interfaces. *J. Phys. D: Appl. Phys.* **47**, 175103 (2014).
3. Nakajima, A., Sumida, Y., Dhyani, M. H., Kawai, H. & Narayanan, E. S. High density two-dimensional hole gas induced by negative polarization at GaN/AlGaN heterointerface. *Appl. Phys. Express* **3**, 121004 (2010).
4. Chaudhuri, R. et al. A polarization-induced 2D hole gas in undoped gallium nitride quantum wells. *Science* **365**, 1454–1457 (2019).
5. Krishna, A., Raj, A., Hatui, N., Keller, S. & Mishra, U. K. Investigation of nitrogen polar p-type doped GaN/$Al_xGa_{(1-x)}N$ superlattices for applications in wide-bandgap p-type field effect transistors. *Appl. Phys. Lett.* **115**, 172105 (2019).
6. Zheng, Z. et al. High I_{on} and I_{on}/I_{off} ratio enhancement-mode buried p-channel GaN MOSFETs on p-GaN gate power HEMT platform. *IEEE Electron Device Lett.* **41**, 26–29 (2020).
7. Raj, A. et al. Demonstration of a GaN/AlGaN superlattice-based p-channel FinFET with high ON-current. *IEEE Electron Device Lett.* **41**, 220–223 (2020).
8. Krishna, A. et al. AlGaN/GaN superlattice-based p-type field-effect transistor with tetramethylammonium hydroxide treatment. *Phys. Status Solidi A* **217**, 1900692 (2020).
9. Nanen, Y., Kato, M., Suda, J. & Kimoto, T. Effects of nitridation on 4H-SiC MOSFETs fabricated on various crystal faces. *IEEE Trans. Electron Devices* **60**, 1260–1262 (2013).
10. Hatakeyama, T. et al. Characterization of traps at nitrided SiO_2/SiC interfaces near the conduction band edge by using Hall effect measurements. *Appl. Phys. Express* **10**, 046601 (2017).

11. Okamoto, M., Tanaka, M., Yatsuo, T. & Fukuda, K. Effect of the oxidation process on the electrical characteristics of 4H-SiC p-channel metal-oxide-semiconductor field-effect transistors. *Appl. Phys. Lett.* **89**, 023502 (2006).
12. Noborio, M., Suda, J. & Kimoto, T. P-channel mosfets on 4H-SiC {0001} and nonbasal faces fabricated by oxide deposition and N_2O annealing. *IEEE Trans. Electron Devices* **56**, 1953–1958 (2009).
13. Pearton, S., Ren, F., Tadjer, M. & Kim, J. Perspective: Ga_2O_3 for ultra-high power rectifiers and MOSFETS. *J. Appl. Phys.* **124**, 220901 (2018).
14. Isberg, J. et al. High carrier mobility in single-crystal plasma-deposited diamond. *Science* **297**, 1670–1672 (2002).
15. Pernot, J. et al. Hall hole mobility in boron-doped homoepitaxial diamond. *Phys. Rev. B* **81**, 205203 (2010).
16. Akimoto, I., Handa, Y., Fukai, K. & Naka, N. High carrier mobility in ultrapure diamond measured by time-resolved cyclotron resonance. *Appl. Phys. Lett.* **105**, 032102 (2014).
17. Koizumi, S., Umezawa, H., Pernot, J. & Suzuki, M. (eds) *Power Electronics Device Applications of Diamond Semiconductors* (Woodhead Publishing, 2018).
18. Geis, M. W. et al. Progress toward diamond power field-effect transistors. *Phys. Status Solidi A* **215**, 1800681 (2018).
19. Donato, N., Rouger, N., Pernot, J., Longobardi, G. & Udrea, F. Diamond power devices: state of the art, modelling, figures of merit and future perspective. *J. Phys. D: Appl. Phys.* **53**, 093001 (2020).
20. Maier, F., Riedel, M., Mantel, B., Ristein, J. & Ley, L. Origin of surface conductivity in diamond. *Phys. Rev. Lett.* **85**, 3472–3475 (2000).
21. Strobel, P., Riedel, M., Ristein, J. & Ley, L. Surface transfer doping of diamond. *Nature* **430**, 439–441 (2004).
22. Verona, C. et al. V_2O_5 MISFETs on H-terminated diamond. *IEEE Trans. Electron Devices* **63**, 4647–4653 (2016).
23. Kawarada, H. et al. Durability-enhanced two-dimensional hole gas of C-H diamond surface for complementary power inverter applications. *Sci. Rep.* **7**, 42368 (2017).
24. Ren, Z. et al. Diamond field effect transistors with MoO_3 gate dielectric. *IEEE Electron Device Lett.* **38**, 786–789 (2017).
25. Kasu, M. Diamond field-effect transistors for RF power electronics: novel NO_2 hole doping and low-temperature deposited Al_2O_3 passivation. *Jpn. J. Appl. Phys.* **56**, 01AA01 (2017).
26. Yu, X. et al. A high frequency hydrogen-terminated diamond MISFET with f_T/f_{max} of 70/80 GHz. *IEEE Electron Device Lett.* **39**, 1373–1376 (2018).
27. Yin, Z. et al. Enhanced transport in transistor by tuning transition-metal oxide electronic states interfaced with diamond. *Sci. Adv.* **4**, eaau0480 (2018).
28. Geis, M. et al. Stable, low-resistance, 1.5 to 3.5 kΩ sq^{-1}, diamond surface conduction with a mixed metal-oxide protective film. *Diam. Relat. Mater.* **106**, 107819 (2020).
29. Li, Y. et al. Mobility of two-dimensional hole gas in H-terminated diamond. *Phys. Status Solidi RRL* **12**, 1700401 (2018).
30. Sasama, Y. et al. Charge-carrier mobility in hydrogen-terminated diamond field-effect transistors. *J. Appl. Phys.* **127**, 185707 (2020).
31. Sasama, Y. et al. High-mobility diamond field effect transistor with a monocrystalline h-BN gate dielectric. *APL Mater.* **6**, 111105 (2018).
32. Sasama, Y. et al. Quantum oscillations in diamond field-effect transistors with a h-BN gate dielectric. *Phys. Rev. Mater.* **3**, 121601(R) (2019).
33. Zibrov, A. A. et al. Tunable interacting composite fermion phases in a half-filled bilayer-graphene Landau level. *Nature* **549**, 360–364 (2017).
34. Castellanos-Gomez, A. et al. Deterministic transfer of two-dimensional materials by all-dry viscoelastic stamping. *2D Mater.* **1**, 011002 (2014).
35. Sque, S. J., Jones, R. & Briddon, P. R. Structure, electronics, and interaction of hydrogen and oxygen on diamond surfaces. *Phys. Rev. B* **73**, 085313 (2006).
36. Stepanov, P. et al. Untying the insulating and superconducting orders in magic-angle graphene. *Nature* **583**, 375–378 (2020).
37. Young, A. et al. Electronic compressibility of layer-polarized bilayer graphene. *Phys. Rev. B* **85**, 235458 (2012).
38. Kitabayashi, Y. et al. Normally-off C-H diamond MOSFETs with partial C-O channel achieving 2-kV breakdown voltage. *IEEE Electron Device Lett.* **38**, 363–366 (2017).
39. Zhang, J.-F. et al. Characterization and mobility analysis of normally off hydrogen-terminated diamond metal–oxide–semiconductor field-effect transistors. *Phys. Status Solidi A* **217**, 1900462 (2020).
40. Liu, J. W., Oosato, H., Liao, M. Y. & Koide, Y. Enhancement-mode hydrogenated diamond metal-oxide-semiconductor field-effect transistors with Y_2O_3 oxide insulator grown by electron beam evaporator. *Appl. Phys. Lett.* **110**, 203502 (2017).
41. Oi, N. et al. Normally-off two-dimensional hole gas diamond MOSFETs through nitrogen-ion implantation. *IEEE Electron Device Lett.* **40**, 933–936 (2019).
42. Tordjman, M., Weinfeld, K. & Kalish, R. Boosting surface charge-transfer doping efficiency and robustness of diamond with WO_3 and ReO_3. *Appl. Phys. Lett.* **111**, 111601 (2017).
43. Verona, C. et al. Influence of surface crystal-orientation on transfer doping of V_2O_5/H-terminated diamond. *Appl. Phys. Lett.* **112**, 181602 (2018).
44. Wang, Y.-F. et al. Normally-off hydrogen-terminated diamond field-effect transistor with Al_2O_3 dielectric layer formed by thermal oxidation of Al. *Diam. Relat. Mater.* **81**, 113–117 (2018).
45. Ren, Z. et al. High performance single crystalline diamond normally-off field effect transistors. *IEEE J. Electron Devices Soc.* **7**, 82–87 (2018).
46. Fei, W., Bi, T., Iwataki, M., Imanishi, S. & Kawarada, H. Oxidized Si terminated diamond and its MOSFET operation with SiO_2 gate insulator. *Appl. Phys. Lett.* **116**, 212103 (2020).
47. Sze, S. M. and Ng, K. K. *Physics of Semiconductor Devices* 3rd edn (John Wiley & Sons, 2007).
48. Saha, N. C. & Kasu, M. Heterointerface properties of diamond MOS structures studied using capacitance–voltage and conductance–frequency measurements. *Diam. Relat. Mater.* **91**, 219–224 (2019).
49. Matsumoto, T. et al. Inversion channel mobility and interface state density of diamond MOSFET using N-type body with various phosphorus concentrations. *Appl. Phys. Lett.* **114**, 242101 (2019).
50. Zhang, X. et al. Insight into Al_2O_3/B-doped diamond interface states with high-temperature conductance method. *Appl. Phys. Lett.* **117**, 092104 (2020).
51. Peterson, R. et al. Analysis of mobility-limiting mechanisms of the two-dimensional hole gas on hydrogen-terminated diamond. *Phys. Rev. B* **102**, 075303 (2020).
52. Takahide, Y. et al. Quantum oscillations of the two-dimensional hole gas at atomically flat diamond surfaces. *Phys. Rev. B* **89**, 235304 (2014).
53. Davies, J. H. *The Physics of Low-Dimensional Semiconductors: An Introduction* (Cambridge Univ. Press, 1998).
54. Zhang, J. et al. Point defects in two-dimensional hexagonal boron nitride: a perspective. *J. Appl. Phys.* **128**, 100902 (2020).
55. Stacey, A. et al. Evidence for primal sp^2 defects at the diamond surface: candidates for electron trapping and noise sources. *Adv. Mater. Interfaces* **6**, 1801449 (2019).
56. Tokuda, N. et al. Atomically flat diamond (111) surface formation by homoepitaxial lateral growth. *Diam. Relat. Mater.* **17**, 1051–1054 (2008).
57. Daligou, G. & Pernot, J. 2D hole gas mobility at diamond/insulator interface. *Appl. Phys. Lett.* **116**, 162105 (2020).
58. Kawarada, H. High-current metal oxide semiconductor field-effect transistors on H-terminated diamond surfaces and their high-frequency operation. *Jpn. J. Appl. Phys.* **51**, 090111 (2012).
59. Yamaguchi, T. et al. Low-temperature transport properties of holes introduced by ionic liquid gating in hydrogen-terminated diamond surfaces. *J. Phys. Soc. Jpn* **82**, 074718 (2013).
60. Macheda, F. & Bonini, N. Magnetotransport phenomena in p-doped diamond from first principles. *Phys. Rev. B* **98**, 201201(R) (2018).

Acknowledgements

We thank H. Osato, E. Watanabe, D. Tsuya, Y. Nishimiya and F. Uesugi for their technical support. We also thank J. Inoue for helpful discussions, and T. Ando, S. Koizumi, T. Teraji, Y. Wakayama and T. Nakayama for their support. This study was financially supported by JSPS KAKENHI (grant nos. JP19J12696 (Y.S.), JP19H05790 (K.W. and T.T.), JP20H00354 (K.W. and T.T.), JP19H02605 (Y.T.) and JP25287093 (Y.T.)); the Elemental Strategy Initiative (grant no. JPMXP0112101001 (K.W. and T.T.)); and the Nanotechnology Platform Project (Y.S. and Y.T.) of MEXT, Japan.

Author contributions

Y.T. and Y.S. conceived the project. Y.S. fabricated the devices and performed the electrical characterizations. Y.S. and Y.T. performed the data analysis and modelling. T.K., M.I. and Y.T. assisted in fabricating the devices. K.W. and T.T. grew the hBN crystals. Y.S., T.U. and Y.T. wrote the manuscript. All the authors discussed the results and commented on the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41928-021-00689-4.

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Peer review information *Nature Electronics* thanks Nazareno Donato, Moshe Tordjman and Cui Yu for their contribution to the peer review of this work.

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